IE2024 - Probability & Statistics

Introduction to Data Analytics and Sampling

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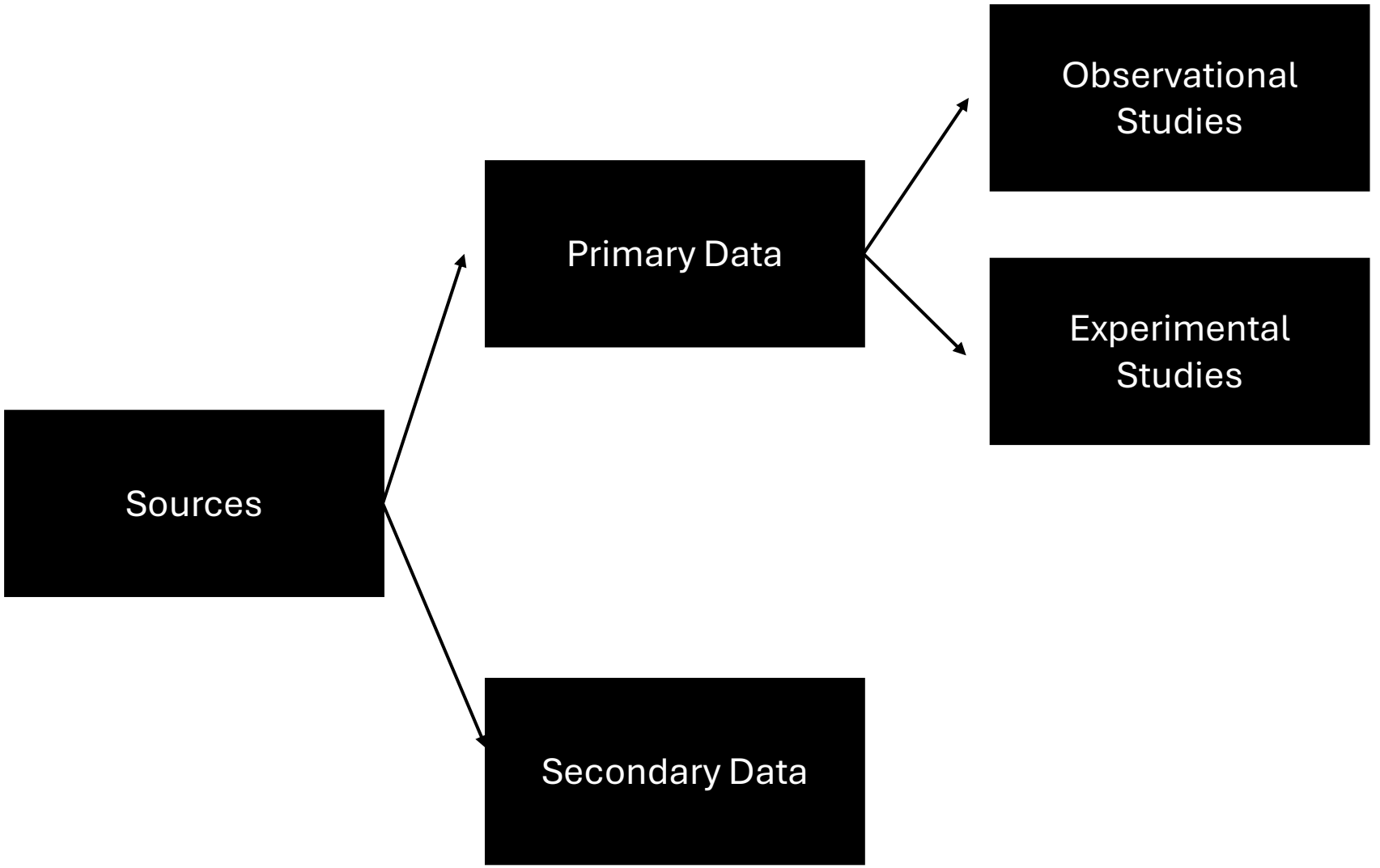


Why is Statistics?

- Statistics is the study of uncertainty.
- We need statistics to identify the variability in data.



Sources of Data



Primary Data

- Observational Studies
  - Data is collected by observing subjects in a natural or controlled environment without interfering.
  - The researcher does not manipulate variables.
  - Example: Watching consumer behavior in a shopping mall.
- Experimental Studies
  - Data is collected by manipulating one or more variables to observe changes in outcomes.
  - A controlled experiment is conducted to determine cause-and-effect relationships.
  - Example: Testing the effect of a new drug on patients by giving the drug to one group and a placebo to another.

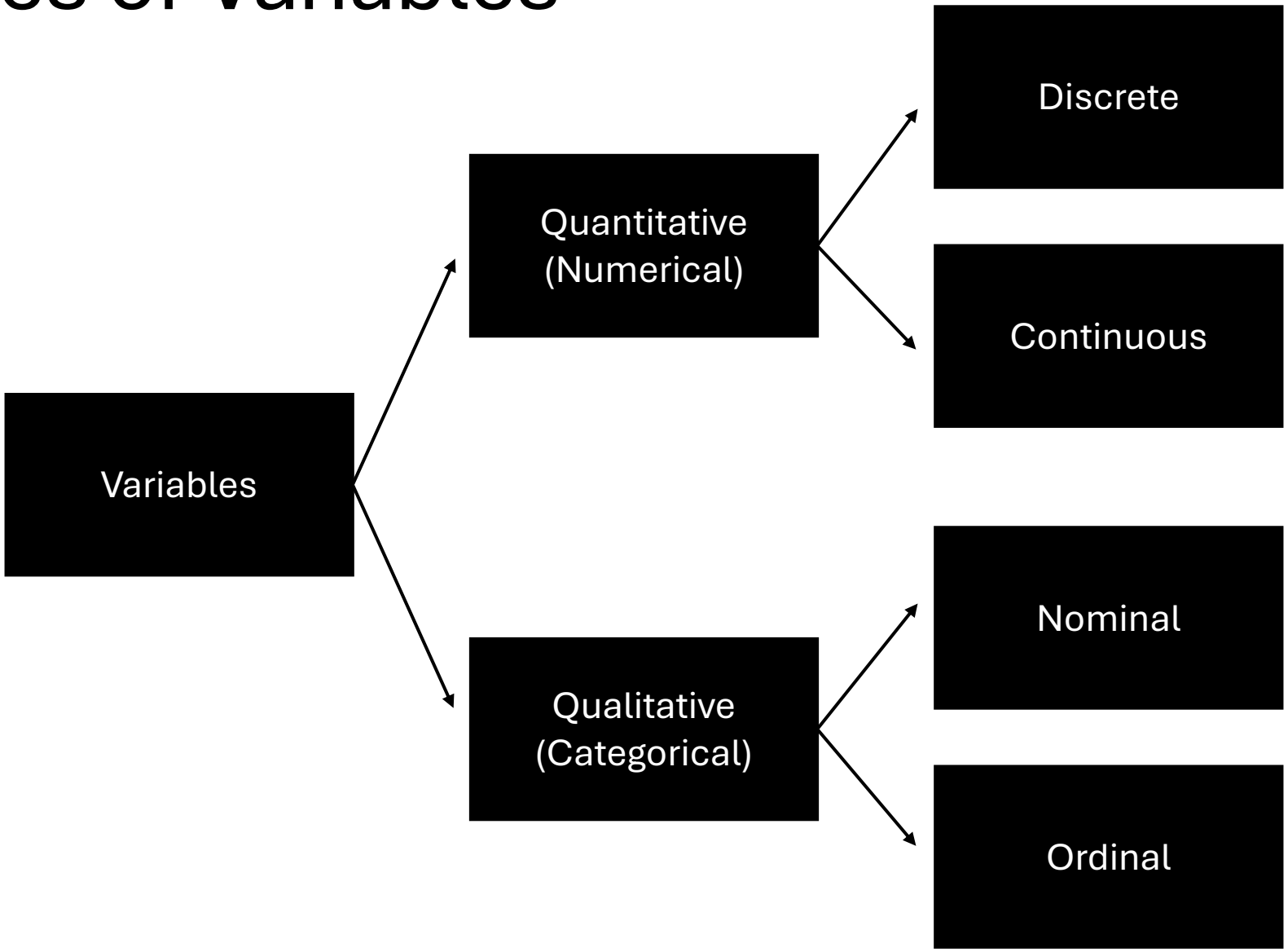
Secondary Data

- Secondary data is information that has already been collected and published by another person or organization. It is used for a different purpose than originally intended.
- Examples:
  - Government census data
  - Research papers
  - Company financial reports
  - Online databases

Variable

- Variable is a characteristic/ property of a person/ a living thing/ an item.
- Ex- Age, Gender, Temperature
- The values of a variable is called as data.
- This value may be a number, a word or a symbol.

Types of Variables



Types of Variables

- **Quantitative Variables** - Quantitative variables are numerical and can be measured. They represent quantities and can be subjected to arithmetic operations.
- **Discrete Variables**: These take on a finite number of values, often counted in whole numbers.
  - Examples: Number of students in a class, number of cars in a parking lot.
- **Continuous Variables**: These can take on an infinite number of values within a given range and are often measured.
  - Examples: Height, weight, temperature, time.

Types of Variables

- Based on the measurement scale, quantitative variables can further be classified into two areas.
- **Interval Scale** - An interval scale is a type of quantitative measurement scale where the intervals between values are meaningful and equal.
- However, an interval scale does not have a true zero point.
- Characteristics:
  - Equal intervals between values.
  - No true zero point (zero does not indicate the absence of the quantity).
- Examples:
  - Temperature in Celsius or Fahrenheit (the difference between 20°C and 30°C is the same as the difference between 30°C and 40°C, but 0°C does not mean "no temperature").
  - IQ scores.

Types of Variables

- **Ratio Scale** - A ratio scale is a type of quantitative measurement scale that has all the characteristics of an interval scale, but also includes a true zero point.
- This means that zero on a ratio scale indicates the absence of the quantity being measured.
- Characteristics:
  - Equal intervals between values.
  - True zero point (zero indicates the absence of the quantity).
- Examples:
  - Weight (0 kg means no weight).
  - Height (0 cm means no height).
  - Income (0 dollars means no income).
  - Number of items (0 items means no items).

Types of Variables

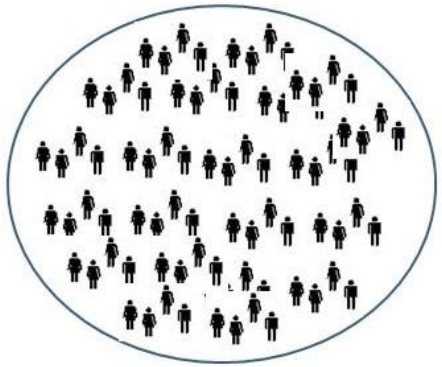
- **Qualitative Variables** - Qualitative variables, also known as categorical variables, describe qualities or characteristics and cannot be measured numerically.
- **Nominal Variables**: These categories do not have a meaningful order or ranking.
  - Examples: Gender, nationality, type of car.
- **Ordinal Variables**: These categories have a meaningful order or ranking, but the intervals between the categories are not necessarily equal.
  - Examples: Education level (high school, bachelor's, master's, PhD), satisfaction rating (satisfied, neutral, dissatisfied).

4 Main Domains of Data Analytics

- Descriptive Analytics
  - To summarize and describe the main features of a dataset.
  - Explains the past and current state.
    - Numerical Summaries
    - Data Visualization
- Diagnostic Analytics
  - To determine the causes of observed phenomena or patterns in the data.
    - Statistical Inference
- Predictive Analytics
  - To predict future and unseen present outcomes based on historical data.
    - Regression Analysis
- Prescriptive Analytics

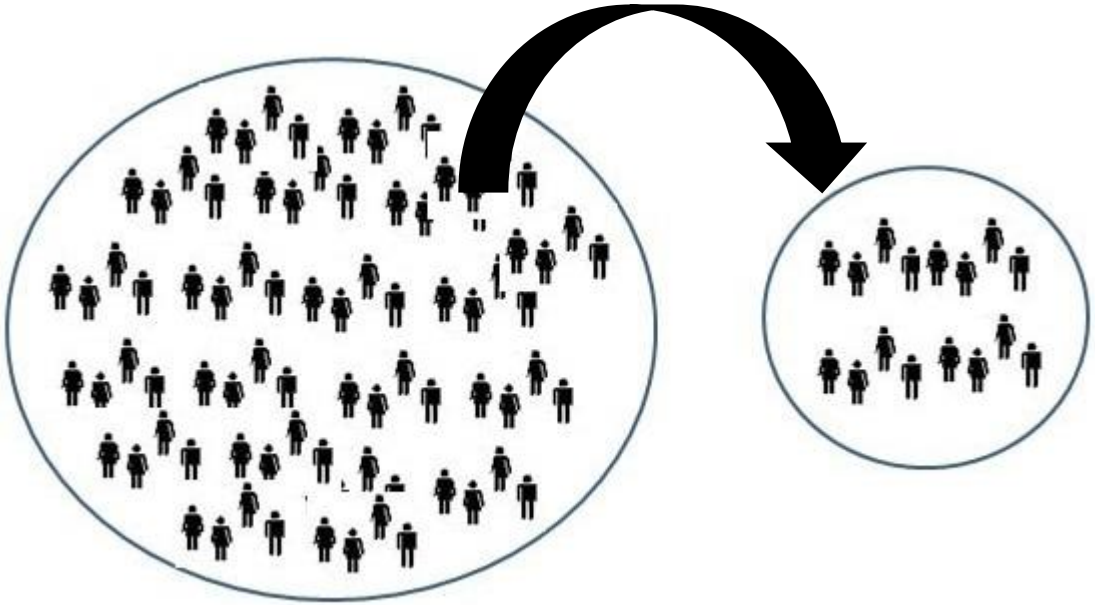
Population

- In statistics, a population is the entire pool in which a specific matter is interested to be checked, or a specific experiment is interested to be done.
- A quantity of value that describes the population is called a Parameter
- Ex- Population mean, Population proportion, Population variance



Sample

- In statistics, a sample is a subset of the population in which a specific matter is going to be checked, or a specific experiment is going to be done.

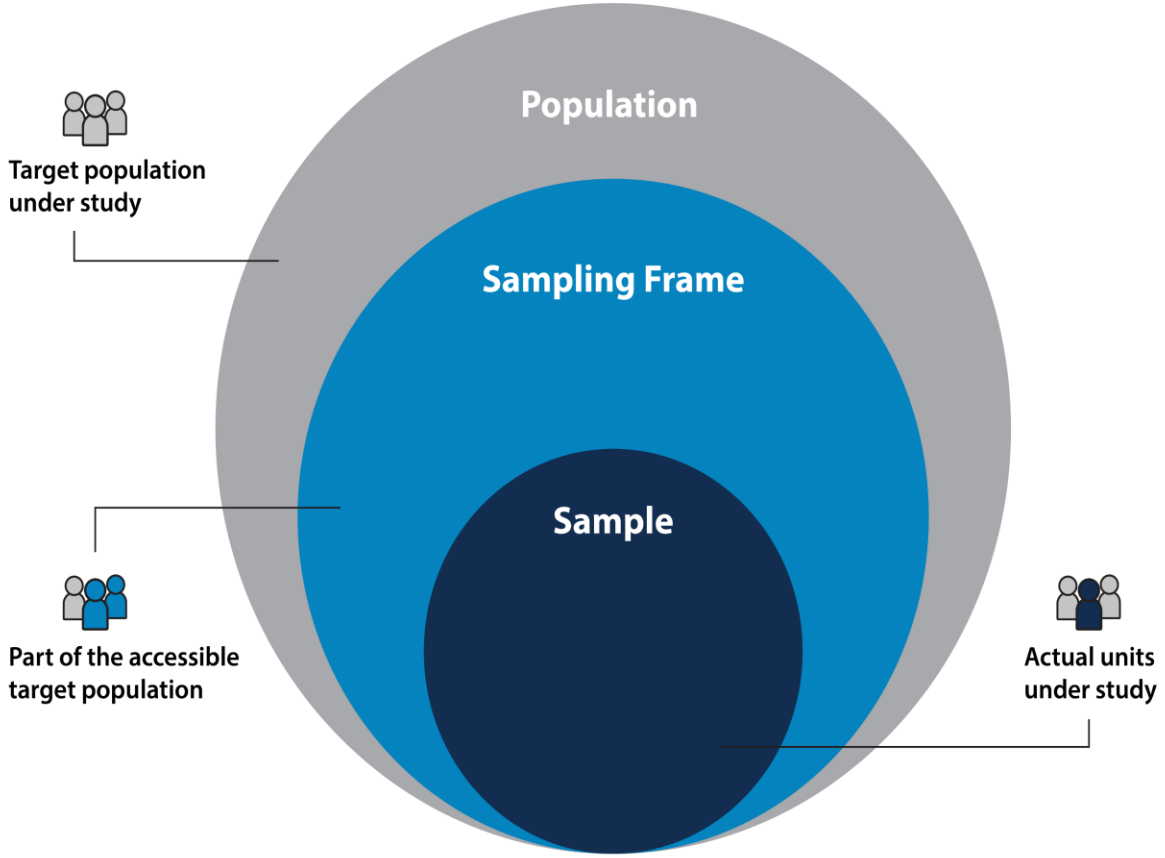


Probability Sampling

- Each sample taken from the population should be a random sample.
- The process of taking these random samples is specially called as **Probability Sampling**.

Sampling Frame

- A sampling frame is a list or database containing the elements from which a sample is drawn for a probability sampling procedure.



Non-Probability Sampling

- But sometimes, because of unavoidable reasons, taking random samples may be difficult to proceeded.
- In those cases, there is another technique called **Non-probability Sampling** can be used.
- This involves non-random selection based on convenience or other criteria, allowing you to easily collect data.

Non-Probability Sampling

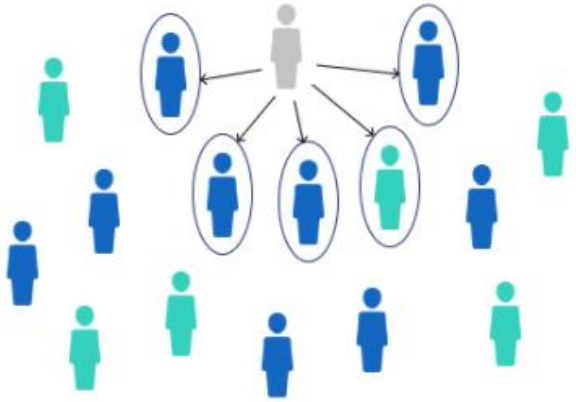
- In a non-probability sample, individuals are selected based on non-random criteria, and not every individual has a chance of being included.
- This type of sample is easier and cheaper to access, but it has a higher risk of sampling bias.
- That means the inferences you can make about the population are weaker than with probability samples, and your conclusions may be more limited.
- If you use a non-probability sample, you should still aim to make it as representative of the population as possible.
- Non-probability sampling techniques are often used in exploratory and qualitative researches.
- In these types of research, the aim is not to test a hypothesis about a broad population, but to develop an initial understanding of a small or under-researched population.

Non-Probability Sampling

- Non-Probability Sampling methods:
  - Convenience Sampling
  - Voluntary Response Sampling
  - Purposive Sampling (Judgement Sampling)
  - Snowball Sampling

Convenience Sampling

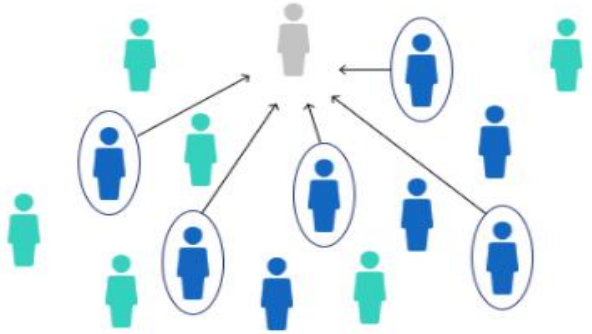
- A convenience sample simply includes the individuals who happen to be most accessible to the researcher.



- This is an easy and inexpensive way to gather initial data, but there is no way to tell if the sample is representative of the population, so it can't produce generalizable results.
- Ex:-You are researching opinions about student support services in your university, so after each of your classes, you ask your fellow students to complete a survey on the topic. This is a convenient way to gather data, but as you only surveyed students taking the same classes as you at the same level, the sample is not representative of all the students at your university.

Voluntary Response Sampling

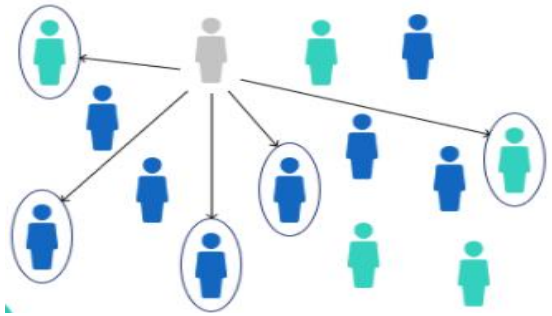
- A voluntary response sample is mainly based on ease of access. Instead of the researcher choosing participants and directly contacting them, people volunteer themselves (e.g. by responding to a public online survey).



- Voluntary response samples are always at least somewhat biased, as some people will inherently be more likely to volunteer than others.
- Ex:-You send out the survey to all students at your university and a lot of students decide to complete it. This can certainly give you some insight into the topic, but the people who responded are more likely to be those who have strong opinions about the student support services, so you can't be sure that their opinions are representative of all students.

Purposive Sampling (Judgement Sampling)

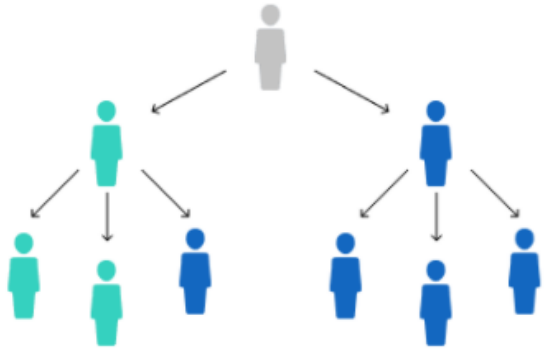
- This type of sampling, also known as judgement sampling, involves the researcher using their expertise to select a sample that is most useful to the purposes of the research.



- It is often used in qualitative research, where the researcher wants to gain detailed knowledge about a specific phenomenon rather than make statistical inferences, or where the population is very small and specific. An effective purposive sample must have clear criteria and rationale for inclusion.
- Ex:-You want to know more about the opinions and experiences of disabled students at your university, so you purposefully select a number of students with different support needs in order to gather a varied range of data on their experiences with student services.

Snowball Sampling

- If the population is hard to access, snowball sampling can be used to recruit participants via other participants.
- The number of people you have access to “snowballs” as you get in contact with more people.



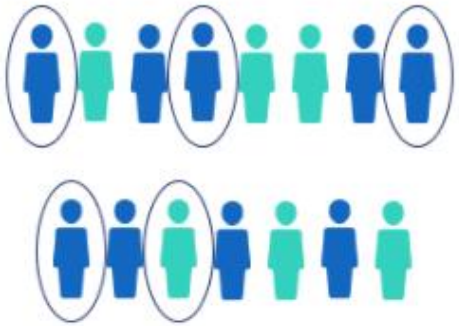
- Ex:-You are researching experiences of homelessness in your city. Since there is no list of all homeless people in the city, probability sampling isn’t possible. You meet one person who agrees to participate in the research, and she puts you in contact with other homeless people that she knows in the area.

Probability Sampling

- Probability sampling means that every member of the population has a chance of being selected.
- It is mainly used in quantitative research.
- If you want to produce results that are representative of the whole population, probability sampling techniques are the most valid choice.
- These are normally applied for quantitative researches.
  - Simple Random Sampling
  - Stratified Sampling
  - Cluster Sampling

Simple Random Sampling

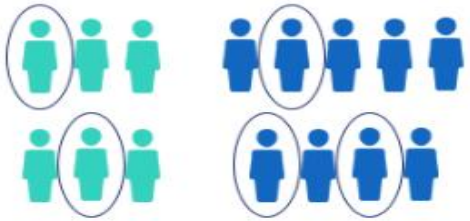
- In a simple random sample, every member of the population has an equal chance of being selected.
- Your sampling frame should include the whole population.



- To conduct this type of sampling, you can use tools like random number generators or other techniques that are based entirely on chance.
- Ex:-You want to select a simple random sample of 100 employees of Company X. You assign a number to every employee in the company database from 1 to 1000, and use a random number generator to select 100 numbers.

Stratified Random Sampling

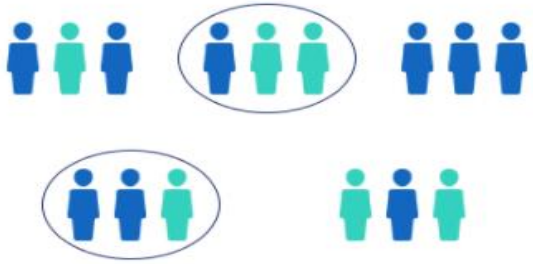
- Stratified sampling involves dividing the population into subpopulations that may differ in important ways.
- It allows you draw more precise conclusions by ensuring that every subgroup is properly represented in the sample.
- To use this sampling method, you divide the population into subgroups (called strata) based on the relevant characteristic (e.g. gender, age range, income bracket, job role).
- Then Simple Random Sampling or Systematic Sampling can be used for getting samples from each strata.



- Ex:-The company has 800 female employees and 200 male employees. You want to ensure that the sample reflects the gender balance of the company, so you sort the population into two strata based on gender. Then you use random sampling on each group, selecting 80 women and 20 men, which gives you a representative sample of 100 people.

Cluster Sampling

- Cluster sampling also involves dividing the population into subgroups, but each subgroup should have similar characteristics to the whole sample.
- Instead of sampling individuals from each subgroup, you randomly select entire subgroups.
- Usually, these clusters are created with naturally created groups.



- Ex:-The company has offices in 10 cities across the country (all with roughly the same number of employees in similar roles). You don't have the capacity to travel to every office to collect your data, so you use random sampling to select 3 offices – these are your clusters.

Sample Size

- A good sample size really depends on the context and goals of the research.
- In general, a good sample size is one that accurately represents the population and allows for reliable statistical analysis.
- One of the standard method is Krejcie and Morgan Table

Krejcie and Morgan Table for Sample Size

N	*S*	*N*	*S*	*N*	*S*
10	10	220	140	1200	291
15	14	230	144	1300	297
20	19	240	148	1400	302
25	24	250	152	1500	306
30	28	260	155	1600	310
35	32	270	159	1700	313
40	36	280	162	1800	317
45	40	290	165	1900	320
50	44	300	169	2000	322
55	48	320	175	2200	327
60	52	340	181	2400	331
65	56	360	186	2600	335
70	59	380	191	2800	338
75	63	400	196	3000	341
80	66	420	201	3500	346
85	70	440	205	4000	351
90	73	460	210	4500	354
95	76	480	214	5000	357
100	80	500	217	6000	361
110	86	550	226	7000	364
120	92	600	234	8000	367
130	97	650	242	9000	368
140	103	700	248	10000	370
150	108	750	254	15000	375
160	113	800	260	20000	377
170	118	850	265	30000	379
180	123	900	269	40000	380
190	127	950	274	50000	381
200	132	1000	278	75000	382
210	136	1100	285	1000000	384

Note.—*N* is population size. *S* is sample size.

Source: Krejcie & Morgan, 1970